Improving Wall Design with Hill Climbing (and Beyond)

Due: Friday before class (submit on Moodle + be ready to demo)

Overview

In this assignment, you will improve the wall-design algorithm using **hill climbing**.

You will:

- build on your greedy algorithm,
- implement hill climbing to improve wall layouts,
- experiment with different improvement strategies,
- and evaluate your results using the visualizer.

For students who want to go further, you may extend your work using **simulated annealing**.

Part 1: Hill Climbing (Required)

Objective

Take a wall produced by your greedy algorithm and **improve it using hill climbing**.

Hill climbing means:

- start with an existing wall,
- repeatedly make small changes,
- keep only changes that improve the score,
- stop when no improvement is possible.

What You Must Do

For at least **2 different walls**:

1. Generate an initial wall (using your greedy algorithm).
2. Apply hill climbing:

- make small changes to the wall,
- evaluate using the scoring function,
- accept only improvements.

1. Repeat until no further improvement is possible.

Required Improvements

You must implement **at least 3 different types of hill-climbing moves**, such as:

- swap two artworks
- remove an artwork
- replace an artwork with another candidate
- add an artwork (if space allows)
- reposition artworks along the wall

You should experiment with different combinations of these moves.

What to Show in Class

Be ready to demonstrate:

- your starting (greedy) wall
- your improved (hill-climbed) wall
- the difference in score and/or visual quality
- a brief explanation of what changes helped

Part 2: Writeup (Required)

Submit a short writeup (1–2 pages) including:

1. Hill Climbing Approach

- How does your hill climbing work?
- What kinds of moves did you implement?

2. Results

- What improvements did you observe?
- Which moves were most effective?

3. Limitations

- Where does hill climbing get stuck?
- What kinds of walls are still difficult?

Part 3: Extension — Simulated Annealing (Optional, for full credit)

To go beyond hill climbing, implement **simulated annealing**.

This allows your algorithm to:

- occasionally accept worse moves,
- escape local optima,
- explore more of the solution space.

If you implement annealing, include:

- description of your temperature schedule
- how you modified hill climbing to allow worse moves
- comparison with your hill climbing results

Grading Rubric (100 points total)

Hill Climbing Implementation (40 points)

- 20 pts: Basic hill climbing implemented correctly
- 30 pts: Multiple meaningful move types used
- 40 pts: Strong hill climbing with effective improvements

Exploration and Experimentation (20 points)

- 10 pts: Minimal experimentation
- 15 pts: Tried multiple move types
- 20 pts: Thorough exploration with comparisons

Results and Demonstration (20 points)

- 10 pts: Some visible improvement
- 15 pts: Clear improvement over greedy
- 20 pts: Strong and consistent improvements

Writeup Quality (20 points)

- 10 pts: Basic explanation
- 15 pts: Clear explanation and results
- 20 pts: Insightful analysis and reflection

Simulated Annealing Bonus (up to +20 points)

- +10 pts: Basic annealing implemented
- +15 pts: Effective annealing with good comparison
- +20 pts: Strong implementation with clear improvement over hill climbing

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Important Note on Grading

- The maximum score achievable using **hill climbing alone is 80%**.
- To earn a score above 80%, you must implement **simulated annealing** or an equivalent advanced strategy.

Summary

- Build a greedy wall
- Improve it using hill climbing
- Try multiple move strategies
- Analyze results
- (Optional) Extend using simulated annealing

Key Idea

> *Greedy builds.*
> *Hill climbing improves.*
> *Annealing explores.*

This assignment is about learning how to move between those ideas.

Admin

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